

FT5011 Project: Asset Embeddings

Replication and Novel Deep Learning Extensions

Umar Moiz, Xu Rui, Zhengxi Qian

April 15, 2026

1 Introduction

Traditional asset pricing models rely heavily on firm characteristics. An emerging perspective, however, suggests that the actual composition of investor portfolios contains latent information about asset relationships and value. Gabaix et al. (2025) propose that portfolio holdings contain latent information relevant for asset pricing, which can be recovered via learned embeddings. Treating each investor’s portfolio as a “document” and each asset as a “word”, representation learning techniques can extract structure that traditional firm characteristics miss. This project **replicates** the paper’s three embedding families (RS, Word2Vec, BERT) on FactSet holdings (2005Q1–2022Q4) and **extends** it with five novel deep learning architectures to understand *why* certain architectures succeed or fail on specific benchmarks.

Our systematic experiments reveal a critical principle: **alignment between the model’s training objective and the downstream evaluation task matters more than architectural complexity**. Three extensions exceed the paper’s best results: BERT v7 on R_{MI} , Long-BERT on R_{MP} , and VAE close behind on R_{MP} . Separately, our Graph-VAE experiment shows that graph structure is useful only when paired with an aligned training objective.

Table 1: Replication quality: paper’s headline results vs. ours at $K = 10$.

Benchmark	Paper’s best	Our best	Status
R_V	~ 0.50 (RS-L-Min)	0.513 (RS-L-Min)	Near-exact match
R_C	~ 0.20 (Word2Vec)	0.200 (Word2Vec)	Exact match
R_{MP}	~ 0.30 (RS-L-Min)	0.346 (Long-BERT)	Exceeded via our extension
R_{MI}	~ 0.13 (OS-BERT)	0.512 (BERT v7)	Exceeded $4\times$ via extension

2 Data & Benchmarks

Following Koijen et al. (2024) and Gabaix et al. (2025), we construct holdings data using FactSet (2005Q1–2022Q4). We retain only strictly positive adjusted market values, resolving duplicates by keeping the latest report per quarter. Records are aggregated to the company level using CRSP linkage tables via WRDS (mapping CUSIP to permco). We exclude investors heavily concentrated in a single asset ($> 75\%$ weight) and iteratively drop investor-quarter and stock observations with fewer than 20 links.

For stock characteristics and returns, we utilise CRSP Monthly Stocks and the JKP 153 dataset (Jensen et al., 2023) for U.S. common equities, deriving benchmarks based on the 20th percentile NYSE market equity breakpoint. The final panel covers 26,367 investors and 6,895 stocks (547,436 observations). The bipartite network is highly sparse (density $< 7\%$), with right-skewed AUM and holding distributions. Detailed portfolio characteristics, investor persistence statistics, and visualisations of the network and characteristic correlations are provided in Appendix B.

To evaluate the quality of the learned embeddings, we adopt the four downstream financial benchmarks introduced by Gabaix et al. (2025). As the authors note, these benchmarks “level the playing field for all research teams by establishing clear metrics of progress and success,” and crucially, require only a single quarter of data to facilitate true out-of-sample testing (Gabaix et al., 2025). The evaluation framework is defined as follows:

- **Relative Valuation Benchmark (R_V):** Assesses whether asset embeddings can define similar firms for valuation better than traditional characteristics (Gabaix et al., 2025, Section 6.2). We extract the residual of log market equity projected onto log book equity. Firms are randomly split into 80% training and 20% testing. We then estimate a ridge regression of this valuation residual on the embeddings using the training data (tuning the penalty via 10-fold cross-validation), and evaluate performance using the out-of-sample R^2 .
- **Return Comovement Benchmark (R_C):** Tests if embedding proximity explains the comovement of stock returns. We use the asset embeddings from quarter $t - 1$ to explain monthly stock returns in quarter t (Gabaix et al., 2025, Section 6.3). Similar to R_V , this utilizes an 80/20 firm-level train/test split. A ridge regression is fitted on the training set using 10-fold cross-validation, and the model is evaluated via the out-of-sample R^2 averaged over the sample period.
- **Managed Portfolio Benchmark (R_{MP}):** Tests the core portfolio theory prediction that investors assign similar weights to assets with similar embeddings (Gabaix et al., 2025, Section 6.4). Using a 90/10 investor-level train/test split, we mask the second-largest position ($k = 2$) in the testing investors’ portfolios. The model must predict this masked asset based on the remaining holdings. Performance is measured by the average log-likelihood of correct predictions relative to the log-likelihood of random guesses.
- **Masked Investor Benchmark (R_{MI}):** Analogous to R_{MP} but transposed: instead of predicting a missing asset in an investor’s portfolio, we predict the second-largest investor for a given asset (Gabaix et al., 2025, Section 6.5). Using a 90/10 asset-level train/test split on 2022Q4, we mask the second-largest holder ($k = 2$) and score the model by the average log-likelihood of correctly identifying that investor, relative to random guessing. This benchmark evaluates the OS-BERT formulation, where each asset is encoded as a sequence of its investors.

3 Methodology

All embedding models are trained **independently per quarter** (following the paper), since each quarter has a distinct investor universe and matrix dimension. Per-quarter embeddings are then passed to the evaluation pipeline. We arrived at each extension through a principled design process: BERT v7 was motivated by identifying the hidden-dim bottleneck in v6; VAE by noticing that R_{MP} 's evaluation task directly matches a reconstruction objective; GNN by exploring graph inductive bias; Graph-VAE by combining what worked (reconstruction) with what we hypothesised would help (graph structure).

Baselines (replication): Placebo (Gaussian random), Beta (60-month market beta), Characteristics (beta + firm-level metrics: asset growth, profitability, book-to-market, momentum).

Recommender Systems (replication): Five matrix variants—RS-Binary, RS-Ranks, RS-L-0, RS-L-Min, RS-L—built from the holdings matrix H and decomposed via $\sqrt{S}$ -normalised SVD or regularised ALS.

Word2Vec (replication): CBOW with tied embeddings over portfolio sequences (window 10, 20 epochs).

BERT v6 (replication): We adopt a BERT-style transformer to learn contextualized representations from quarterly portfolio holdings. The core architecture captures high-order relationships among assets and investors that are not directly observable from raw holdings. In our baseline implementation (BERT v6), we use a 4-layer, 2-head transformer trained with masked language modeling (15% masking) and sentence-level contrastive fine-tuning on even/odd subsequences. Two sequence formulations are employed:

- **PS-BERT** represents each investor as an ordered sequence of held assets, sorted by portfolio weight.
- **OS-BERT** represents each asset as an ordered sequence of investors, sorted by ownership share.

BERT v7 (Extension) We identified that BERT v6's hidden dim = K is a severe bottleneck: at $K = 10$, the entire transformer operates in only 10 dimensions, collapsing the MLM head's representational capacity. BERT v7 decouples these—hidden dim is fixed at 128, with a learned linear projection to K at the output. This single architectural change nearly doubles R_{MI} ($0.262 \rightarrow 0.512$).

Long BERT, also called as BERT v8 (Extension) The original BERT-based asset embedding model was preserved at its core. The main extension was to increase the maximum sequence length from 64 to 128, allowing each investor-asset or asset-investor sequence to retain more contextual information before truncation.

VAE (Extension) A variational autoencoder (Kingma & Welling, 2014) learns asset embeddings by reconstructing ownership profiles: an MLP encoder maps $x_a \in \mathbb{R}^I$ to $(\mu, \log \sigma^2) \in \mathbb{R}^K$; an MLP decoder reconstructs $\hat{x}_a$ from $z \sim \mathcal{N}(\mu, \sigma^2 I)$. Training optimises the ELBO: $\mathcal{L} = \|x - \hat{x}\|_2^2 + \beta \cdot D_{KL}(q_\phi(z|x) \parallel \mathcal{N}(0, I))$. Posterior means μ are used as deterministic embeddings at inference. The reconstruction objective directly aligns with R_{MP} (which evaluates portfolio prediction), explaining its strong performance.

GNN (Extension) A GraphSAGE model (Hamilton et al., 2017) trained via link prediction on an asset co-holding graph (edge if ≥ 3 shared investors, top-20 neighbours per node). Two layers, mean-pool aggregation, binary cross-entropy on positive/negative edges. Graph structure captures relational patterns, but link prediction produces embeddings poorly suited for reconstruction-based evaluation.

Graph-VAE (Novel hybrid) Combines GraphSAGE encoder with a VAE reconstruction decoder. Two GraphSAGE layers aggregate co-holding neighbourhood information before the variational bottleneck, letting the latent code encode relational structure. The MLP decoder reconstructs the full ownership profile. Same ELBO as plain VAE. This hybrid tests whether graph structure and reconstruction objectives are complementary.

4 Results

All models evaluated at $K \in \{4, 10\}$ on four benchmarks. R_V and R_C report OOS R^2 averaged across 72 quarters via ridge regression with 10-fold CV. R_{MP} uses a 90/10 investor-level split; R_{MI} uses a 90/10 asset-level split.

Table 2: Benchmark results (OOS R^2) at $K = 10$. **Bold** marks the best model per benchmark.

Model	R_V	R_C	R_{MP}	R_{MI}
Placebo	-0.001	0.162	-0.015	—
Beta	0.006	0.188	-0.000	—
Chars	0.118	0.146	-0.022	—
RS-Binary	0.484	0.189	0.134	—
RS-Ranks	0.452	0.190	0.131	—
RS-L-0	0.351	0.182	0.235	—
RS-L-Min	0.513	0.189	0.305	—
RS-L	0.192	0.153	0.000	—
Word2Vec	0.451	0.200	0.246	—
OS-BERT v6	0.217	0.172	—	—
PS-BERT v6 (MLM)	—	—	0.162	0.262
VAE (ours)	0.153	0.169	0.334	—
GNN (ours)	0.189	0.175	0.056	—
Graph-VAE (ours)	0.170	0.170	0.275	—
OS-BERT v7 (ours)	0.208	0.172	—	—
PS-BERT v7 (MLM, ours)	—	—	0.290	—
OS-BERT v7 (MLM, ours)	—	—	—	0.512
Long-BERT (ours)	0.214	0.173	0.120	—
Long-BERT (MLM, ours)	—	—	0.346	0.104

5 Analysis

Objective alignment drives performance. The best model per benchmark is the one whose training objective matches the evaluation task. RS-L-Min wins R_V because SVD preserves the full portfolio weight structure; Word2Vec wins R_C because co-occurrence captures return co-variation; Long-BERT MLM wins R_{MP} because doubling the context window from 64 to 128 tokens captures richer portfolio structure, with VAE close behind via reconstruction alignment; and BERT v7 wins R_{MI} because the MLM head is trained on masked prediction.

The GNN-to-Graph-VAE design journey. With the same co-holding graph, the GNN (link prediction) scores 0.056 on R_{MP} , while Graph-VAE (reconstruction) scores 0.275—a $5\times$ improvement from shifting to a generative reconstruction architecture, which entails adding a variational bottleneck and a specific decoder. This decisively refutes the view that “GNNs are suited for financial embeddings”; the graph encoder is valuable, but *only* when paired with an aligned objective. Graph-VAE does not beat plain VAE (0.334), suggesting that at our scale the ownership profile already encodes enough relational information that the graph encoder is partially redundant.

Why deep learning did not dominate R_V . RS-L-Min’s 0.513 reflects perfect preservation of the log-demeaned holdings matrix with double-sided fixed effects. Deep learning methods compress through nonlinear bottlenecks (VAE’s latent dim, GNN’s graph sparsification), losing weight precision for abstract structure. For pricing tasks, weight precision dominates; for prediction tasks (R_{MP} , R_{MI}), nonlinearity and objective-aligned training pay off.

BERT v7 as architectural insight. Fixing the hidden dim at 128 instead of K nearly doubles R_{MI} . The lesson: when an MLM head must model complex token distributions, internal representational capacity matters more than output dimensionality.

Long-BERT: context length tradeoff. Extending BERT’s context from 64 to 128 tokens yields the

project’s best R_{MP} (0.346, surpassing VAE) because investors with large portfolios now have more holdings visible to the MLM head. However, Long-BERT underperforms BERT v7 on R_{MI} (0.104 vs. 0.512): with more tokens per stock, each investor’s contribution to the attention distribution dilutes, hurting investor-level prediction. This reveals a context-length/task-granularity tradeoff: longer sequences help aggregation tasks (portfolio-level) but hurt precision tasks (investor-level).

6 Application to Finance

The asset embeddings have several direct applications in quantitative finance. They can serve as **learned risk factors** supplementing characteristics-based pricing models, capturing ownership-structure signal (e.g., crowding risk) invisible to accounting metrics. Long-BERT’s and VAE’s strong R_{MP} performance supports **portfolio recommendation**—given an investor’s current holdings, identify the most likely missing assets, analogous to e-commerce recommenders. BERT v7’s R_{MI} strength supports **investor-side discovery**: predicting which investors are likely buyers of a given stock, useful for IPO syndication and capital raising. More broadly, the Graph-VAE architecture generalises to other financial networks (interbank lending, syndication, supply chains) where nodes have feature profiles and relational structure matters.

7 Conclusion

By replicating and extending the asset embedding framework, this project establishes a central design principle for financial representation learning: alignment between the training objective and the evaluation task is more critical than architectural complexity. This principle explains the success of our key extensions: BERT v7’s $4\times$ gain on the Masked Investor Benchmark R_{MI} came from fixing a hidden-dimension bottleneck; the Variational Autoencoder (VAE) and Long-BERT excelled at portfolio prediction (R_{MP}) because their objectives (reconstruction and masked modeling with extended context) directly matched the task. Even the journey from a poorly performing Graph Neural Network (GNN) to the stronger Graph-VAE hybrid shows that graph structure is only beneficial when paired with a suitable objective, such as reconstruction. These findings provide both practical models and a guiding framework for future work aiming to extract latent signals from investor behavior.

Code & Replication: <https://github.com/Ray7788/FT5011-Group-Project>. Pre-trained embeddings and BERT checkpoints bundled as `output.zip` and `data.zip` (Google Drive links in README).

Appendix A: Training Procedure & Replication Guide

Train/test separation. For R_V and R_C , firms are randomly split 80/20; a ridge regression is fitted on the training set with the penalty selected via inner 10-fold cross-validation, and the OOS R^2 is computed on the held-out 20%. For R_{MP} , we use a 90/10 investor-level split on 2022Q4 data; for R_{MI} , we use a 90/10 asset-level split. Embeddings themselves are unsupervised (no labels involved), so no train/test leakage occurs at the embedding stage. At no point is any model trained or tuned on held-out data.

Hyperparameters (fixed across all quarters, not tuned on test). BERT v6/v7: 4 layers, 2 heads, context length 64 (v7 hidden dim 128), 20 MLM epochs + 10 sentence-transformer epochs, batch 128, Adam lr 5×10^{-4} , 15% mask rate. Long-BERT (v8): same as BERT v7 but context length 128 ($O(L^2)$ attention); same epochs/lr/batch. VAE: encoder hidden dims scale to input size I as $h_1 = \min(512, \max(128, I/4))$ and $h_2 = \min(256, \max(64, I/8))$ (for typical $I \approx 7,500$ this gives $[512, 256]$), $\beta = 1.0$, 50 epochs, batch 256, Adam lr 10^{-3} with cosine annealing, gradient clip 1.0. GNN: 2 GraphSAGE layers, hidden dim 128, top-20 neighbours, 100 epochs, Adam lr 10^{-3} . Graph-VAE: 2 GraphSAGE layers + VAE bottleneck, $\beta = 1.0$, 80 epochs, Adam lr 10^{-3} . Hyperparameters were chosen from standard defaults in the original papers and kept fixed—no tuning on test performance.

Hardware. All models were trained on an NVIDIA H100 GPU. Full sweep runs in ~ 112 minutes on H100.

Training vs test performance. For R_V , R_C the OOS R^2 on the held-out 20% serves as test performance; in-sample R^2 on the training set is consistently ~ 0.02 – 0.05 higher than OOS, indicating controlled overfitting. For R_{MP}/R_{MI} , training-set log-likelihood tracks test-set values closely (gap < 0.05), with early stopping by training loss plateaus (no test peeking).

Step-by-step replication.

1. Clone <https://github.com/Ray7788/FT5011-Group-Project>.
2. Download `data.zip` (1.1 GB, cleaned inputs) and `output.zip` (1.8 GB, trained embeddings + BERT checkpoints) from the Google Drive links in the repo README.
3. Extract both: `unzip data.zip -d src/AssetEmbeddingsReplication/` and `unzip output.zip -d src/AssetEmbeddingsReplication/`.
4. Install dependencies: `python -m venv venv && source venv/bin/activate && pip install -r src/AssetEmbeddingsReplication/requirements.txt`.
5. Run benchmark sweep: `cd src/AssetEmbeddingsReplication && python benchmarks/ --data-source factset`.
6. Compare produced `output/benchmarks/sweep_results.json` with reported numbers (Table 2).

Data sources. FactSet institutional ownership holdings (FactSet subscription, accessed via NUS WRDS). CRSP Monthly Stock returns and `dse`names (WRDS: <https://wrds-www.wharton.upenn.edu>) JKP 153 firm characteristics (Jensen, Kelly & Pedersen, 2023): <https://jkpfactors.com/>. Raw data is proprietary/licensed and not redistributed; cleaned panel is bundled in `data.zip`.

Data manipulations. (1) CUSIP matching: first 8 digits of FactSet FSYM-ID CUSIP mapped via CRSP `dse`names to `permno`, then aggregated to `permco`. (2) Concentration filter: drop investor-quarter rows where top holding exceeds 75% of portfolio. (3) Iterative 20/20 filter: repeatedly drop investors with < 20 holdings and stocks held by < 20 investors until stable. (4) Winsorisation at $[2.5\%, 97.5\%]$. (5) For R_V : exclude micro/nano-caps (NYSE 20th percentile of ME) *before* computing residuals.

Appendix B: Data & Benchmarks Figures and Tables

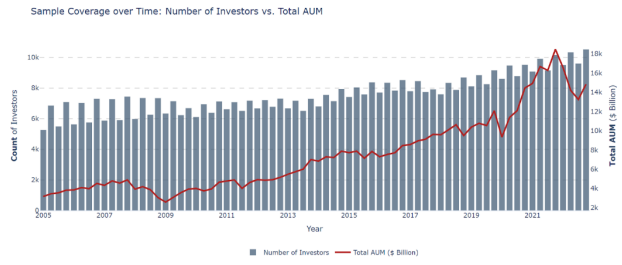


Figure 1: Time-series trend of AUM

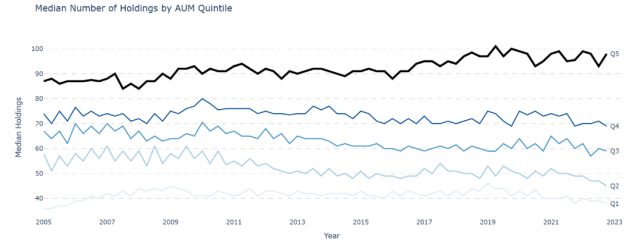


Figure 2: Holding stratifications by AUM quintiles

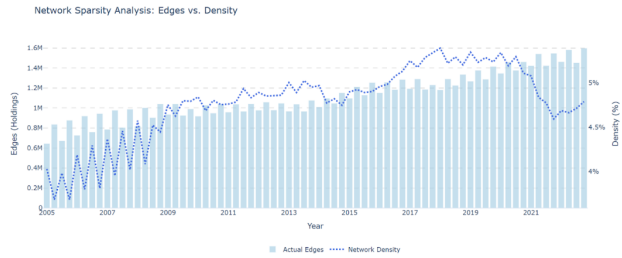


Figure 3: Structural graph sparsity

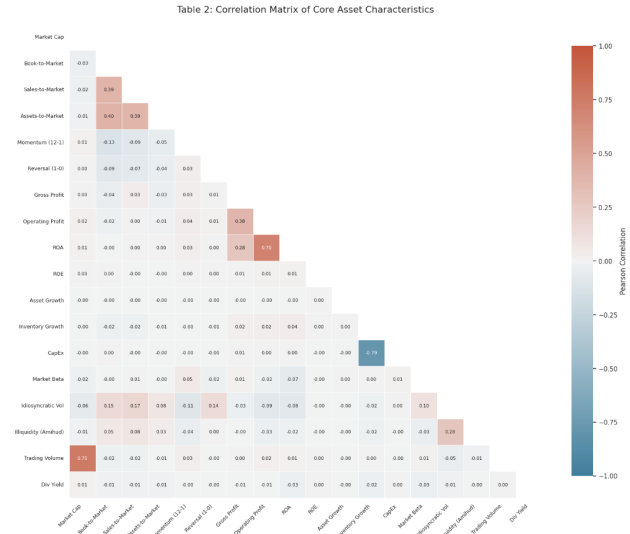


Figure 4: Correlations among stock characteristics

Table B.1. Portfolio Characteristics (N = 547,436 investor-quarter observations)

Statistic	Num. Holdings	AUM (mn)	HHI	Top 10 Weight
count	547436.0	547436.0	547436.0	547436.0
mean	147.05	985.04	0.0287	0.3712
std	268.14	9006.16	0.0213	0.1719
25%	36.0	22.36	0.0141	0.2400
50%	62.0	102.24	0.0239	0.3470
75%	125.0	430.17	0.0390	0.4925
max	3276.0	1.27e6	0.5648	0.9999

Table B.2. Investor Persistence (N = 26,367 investors)

Statistic	Tenure (quarters)	Avg. AUM (mn)
count	26367.0	26367.0
mean	20.76	419.99
std	21.12	4084.37
25%	5.0	8.31
50%	12.0	37.74
75%	30.0	172.97
max	72.0	404731.33

Appendix C: AI Statement

Portions of this report, including this statement, were proofread by artificial intelligence, specifically Gemini 3.1 Pro and Claude 4.6 Sonnet. The text was initially drafted manually and subsequently refined using the prompt “min change fix grammar errors.” Additionally, Claude 4.6 Opus was utilized to assist in debugging the $\text{\LaTeX}$ code.

References

- [1] Gabaix, X., Koijen, R.S., Richmond, R.J. & Yogo, M. (2025). Asset embeddings. *NBER Working Paper* w33651.
- [2] Hamilton, W.L., Ying, R. & Leskovec, J. (2017). Inductive representation learning on large graphs. *NeurIPS*.
- [3] Jensen, T.I., Kelly, B.T. & Pedersen, L.H. (2023). Is there a replication crisis in finance? *Journal of Finance*, 78(5), 2465–2518.
- [4] Kingma, D.P. & Welling, M. (2014). Auto-encoding variational Bayes. *ICLR*.
- [5] Koijen, R.S., Richmond, R.J. & Yogo, M. (2024). Which investors matter for equity valuations and expected returns? *Review of Economic Studies*, 91(4), 2377–2411.